



ENGINEERING MATHEMATICS - II

Dr. G. K. Jagatheswari

Department of Science and Humanities

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UNIT 5 : Fourier series

Session : 4

Sub Topic : Problems on Fourier Series

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Problems on Fourier series

**1. Find the Fourier series expansion of $f(x) = x(1 - x)(2 - x)$ in $(0, 2)$.
Deduce the sum of the series $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \dots$**

Solution:

Given $f(x) = x(1 - x)(2 - x) = x^3 - 3x^2 + 2x$

Here the interval is $(0, 2)$

Comparing with $(0, 2l)$, we have $2l = 2$

$$\therefore l = 1$$

The required Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\pi x + b_n \sin n\pi x)$$

Solution contd...

$$\begin{aligned}a_0 &= \frac{1}{1} \int_0^2 f(x) dx \\&= \int_0^2 x(1-x)(2-x) dx \\&= \int_0^2 (x^3 - 3x^2 + 2x) dx \\&= \left(\frac{x^4}{4} - x^3 + x^2 \right)_0^2 \\a_0 &= 0\end{aligned}$$

Solution contd...

$$\begin{aligned} a_n &= \int_0^2 f(x) \cos \frac{n\pi x}{l} dx \\ &= \int_0^2 x(1-x)(2-x) \cos n\pi x dx \\ &= \int_0^2 (x^3 - 3x^2 + 2x) \cos n\pi x dx \\ &= \left[(x^3 - 3x^2 + 2x) \left(\frac{\sin n\pi x}{n\pi} \right) - (3x^2 - 6x + 2) \left(\frac{-\cos n\pi x}{n^2 \pi^2} \right) \right. \\ &\quad \left. + (6x - 6) \left(\frac{-\sin n\pi x}{n^3 \pi^3} \right) - (6) \left(\frac{\cos n\pi x}{n^4 \pi^4} \right) \right]_0^2 \end{aligned}$$

Solution contd...

$$a_n = \left[\frac{2}{n^2 \pi^2} - \frac{2}{n^2 \pi^2} \right] - 6 \left[\frac{1}{n^4 \pi^4} - \frac{1}{n^4 \pi^4} \right]$$

$$a_n = 0$$

$$\begin{aligned} b_n &= \int_0^2 f(x) \sin \frac{n\pi x}{l} dx \\ &= \int_0^2 x(1-x)(2-x) \sin n\pi x dx \\ &= \int_0^2 (x^3 - 3x^2 + 2x) \sin n\pi x dx \end{aligned}$$

Solution contd..

$$\begin{aligned} b_n &= \left[(x^3 - 3x^2 + 2x) \left(\frac{-\cos n\pi x}{n\pi} \right) - (3x^2 - 6x + 2) \left(\frac{-\sin n\pi x}{n^2\pi^2} \right) \right. \\ &\quad \left. + (6x - 6) \left(\frac{\cos n\pi x}{n^3\pi^3} \right) - (6) \left(\frac{\sin n\pi x}{n^4\pi^4} \right) \right]_0^2 \\ b_n &= [0 - 0] + 6 \left[\frac{1}{n^3\pi^3} + \frac{1}{n^3\pi^3} \right] \end{aligned}$$

$$b_n = \frac{12}{n^3\pi^3}$$

Solution contd...

Hence the required Fourier series is

$$f(x) = x(1 - x)(2 - x) = \sum_{n=1}^{\infty} \left(\frac{12}{n^3 \pi^3} \sin n\pi x \right) \dots\dots(1)$$

Deduction

Put $x = \frac{1}{2}$ in (1)

$$1 - \frac{3}{4} + \frac{1}{8} = \sum_{n=1}^{\infty} \left(\frac{12}{n^3 \pi^3} \sin \frac{n\pi}{2} \right)$$
$$\frac{3}{8 * 12} = \frac{1}{\pi^3} \left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \dots \right]$$
$$\frac{\pi^3}{32} = \left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \dots \right]$$



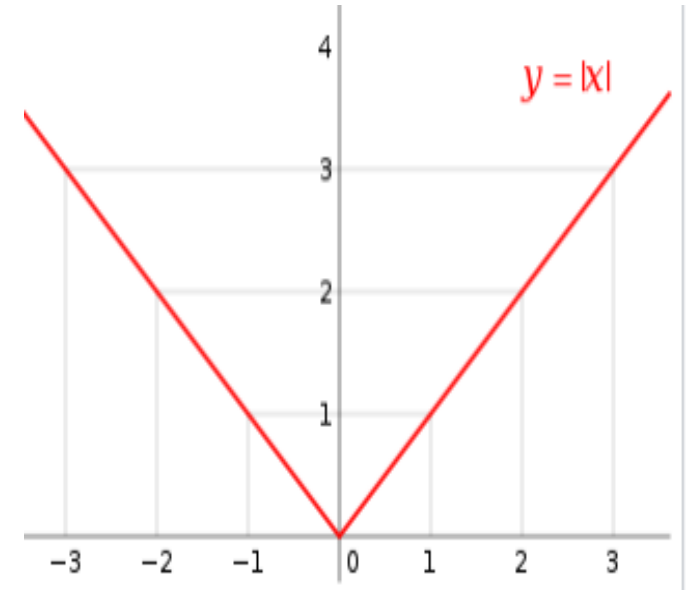
2. Obtain the Fourier Series to represent the function $f(x) = |x|$, $-\pi < x < \pi$.

Solution:

Since $f(x)$ is an even function, the Fourier Series expansion of $f(x)$ is given by

$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$ where

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_0^{\pi} x dx \\ &= \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} \end{aligned}$$



$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Solution Contd....

$$\therefore a_0 = \frac{\pi}{2}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos nx dx$$

$$= \frac{2}{\pi} \left[x \cdot \frac{(\sin nx)}{n} - (1) \cdot \frac{(-\cos nx)}{n^2} \right]_0^{\pi}$$

$$\therefore a_n = \frac{2}{\pi n^2} [(-1)^n - 1]$$

Therefore the required Fourier Series is given by

$$f(x) = |x| = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$|x| = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} [(-1)^n - 1] \cos nx$$



THANK YOU

Dr. G K Jagatheswari

Department of Science & Humanities

jagatheswarigk@pes.edu